

Statistical Learning – Week 06

Impurity Measures, Bagging, Random Forest & MARS

August 28, 2026

1 Properties of Impurity Function Ψ

- (i) symmetric in its arguments
- (ii) Ψ is max when $p_1 = \dots = p_k = 1/k$.

We shall show that for any $(p_1, \dots, p_k)$ with $\sum_{i=1}^k p_i = 1$,

$$\Psi(p_1, \dots, p_k) \leq \Psi(1/k, 1/k, \dots, 1/k)$$

$$\begin{aligned} \Psi(p_1, p_2, \dots, p_k) &= \Psi(p_2, \dots, p_k, p_1) = \Psi(p_3, \dots, p_k, p_1, p_1, p_2) \\ &= \Psi(p_k, p_1, p_2, \dots, p_{k-1}) = \Psi_0, \text{ say} \end{aligned}$$

$$\frac{1}{k} \left[(p_1, p_2, \dots, p_k) + (p_2, p_3, \dots, p_k, p_1) + \dots + (p_k, p_1, \dots, p_{k-1}) \right] = \left(\frac{1}{k}, \frac{1}{k}, \dots, \frac{1}{k} \right)$$

$$\begin{aligned} \Psi\left(\frac{1}{k}, \dots, \frac{1}{k}\right) &= \Psi\left(\frac{1}{k}(\alpha_1 + \dots + \alpha_k)\right) \\ &\geq \frac{1}{k}\Psi(\alpha_1) + \frac{1}{k}\Psi(\alpha_2) + \dots + \frac{1}{k}\Psi(\alpha_k) \\ &= \frac{1}{k}\Psi_0 + \frac{1}{k}\Psi_0 + \dots + \frac{1}{k}\Psi_0 = \Psi_0 = \Psi(p_1, \dots, p_k) \end{aligned}$$

- (iii) Ψ is min when one of the p_i 's is 1 and the rest are 0.

$$\Psi(1, 0, 0, \dots, 0) = \Psi(0, 1, 0, \dots, 0) = \dots = \Psi(0, 0, 0, \dots, 1) = \Psi^*, \text{ say}$$

We need to show that for any $(p_1, \dots, p_k)$ with $\sum_{i=1}^k p_i = 1$,

$$\Psi(p_1, p_2, \dots, p_k) \geq \Psi^*$$

$$(p_1, p_2, \dots, p_k) = p_1(1, 0, \dots, 0) + p_2(0, 1, 0, \dots, 0) + \dots + p_k(0, 0, \dots, 0, 1)$$

Using the concavity, we have

$$\begin{aligned} \Psi(p_1, p_2, \dots, p_k) &\geq p_1\Psi(1, 0, \dots, 0) + p_2\Psi(0, 1, 0, \dots, 0) \\ &\quad + \dots + p_k\Psi(0, 0, \dots, 1) \\ &= p_1\Psi^* + p_2\Psi^* + \dots + p_k\Psi^* = \Psi^* \end{aligned}$$

(iv) For any split of t , $\Delta i(t) \geq 0$.

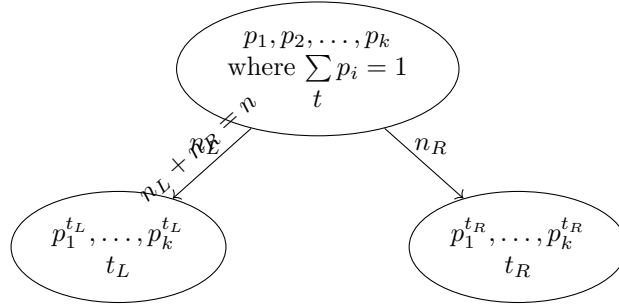


Figure 1: Impurity decrease for a split of node t into t_L and t_R

$$\begin{aligned}
 i(t) &= \Psi(p_1, \dots, p_k) \\
 i(t_L) &= \Psi(p_1^{t_L}, p_2^{t_L}, \dots, p_k^{t_L}) \\
 i(t_R) &= \Psi(p_1^{t_R}, p_2^{t_R}, \dots, p_k^{t_R}) \\
 \Delta i(t) &= \Psi(p_1, \dots, p_k) - \left\{ \frac{n_L}{n} \Psi(p_1^{t_L}, \dots, p_k^{t_L}) + \frac{n_R}{n} \Psi(p_1^{t_R}, \dots, p_k^{t_R}) \right\}
 \end{aligned}$$

We need to show that $\Delta i(t) \geq 0$ for any split.

Note that

$$\begin{aligned}
 p_i &= \frac{n_L}{n} p_i^{t_L} + \frac{n_R}{n} p_i^{t_R} \quad \text{for all } i = 1, 2, \dots, k. \\
 \underline{p} &= \frac{n_L}{n} \underline{p}_L + \frac{n_R}{n} \underline{p}_R \quad \text{where } \frac{n_L}{n} + \frac{n_R}{n} = 1 \\
 &\Rightarrow \Psi(\underline{p}) \geq \frac{n_L}{n} \Psi(\underline{p}_L) + \frac{n_R}{n} \Psi(\underline{p}_R) \\
 &\Rightarrow \Psi(\underline{p}) - \left[\frac{n_L}{n} \Psi(\underline{p}_L) + \frac{n_R}{n} \Psi(\underline{p}_R) \right] \geq 0 \\
 &\quad \underbrace{\Delta i(t)}_{\geq 0}
 \end{aligned}$$

2 Impurity of a Node – Definition and Popular Choices

Any concave function $\Psi(p_1, \dots, p_k)$ which is symmetric in its arguments, (max at uniform, min at vertices) can be used to define the impurity of a node.

Some popular choices are:

Gini Index

i) $\Psi(p_1, p_2, \dots, p_k) = \sum_{i \neq j} p_i p_j$ (This index $\sum_{i \neq j} p_i p_j$ is known as the Ginn's impurity index.

$$\begin{aligned}
 (p_1, p_2, \dots, p_k) &\times ?? \\
 \text{Gini index } \sum_{i \neq j} p_i p_j &\text{ is known as Gini's impurity index.}
 \end{aligned}$$

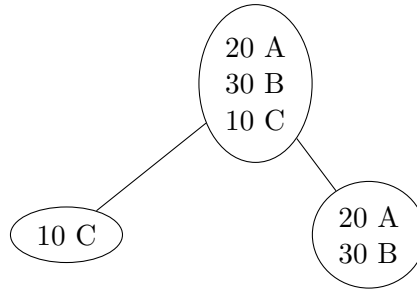


Figure 2: Example: Parent node 20 A, 30 B, 10 C split into pure 10 C node and mixed 20 A 30 B node. $\Psi(p_1, p_2, p_3) = 0$ would indicate purity, but here we have a mixture of observations from two classes.

$$\left(\sum p_i\right)^2 = 1$$

$$\sum p_i^2 + \sum_{i \neq j} p_i p_j = 1 \Rightarrow \sum_{i \neq j} p_i p_j = 1 - \sum_{i=1}^k p_i^2$$

Entropy

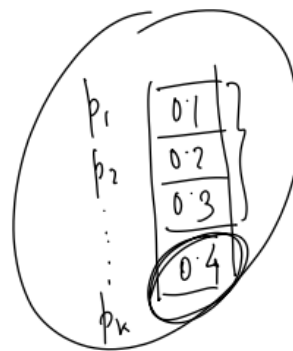
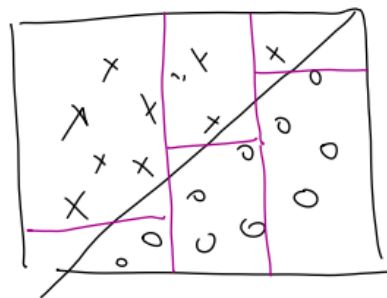
ii) Entropy

$$\Psi(p_1, p_2, \dots, p_k) = - \sum_{i=1}^k p_i \log p_i$$

Misclassification Impurity

iii) Misclassification impurity

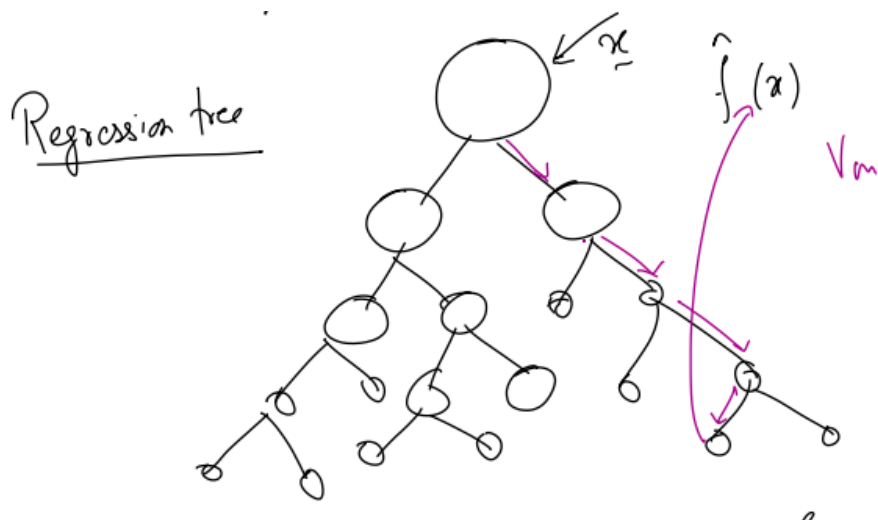
$$\Psi(p_1, \dots, p_k) = 1 - \max\{p_1, \dots, p_k\}$$



3 To Reduce Variability – Motivation for Ensembles

To reduce the variability of the regression tree / classification tree methods, often we construct several trees and aggregate the results.

Regression tree: $\hat{f}(x)$



$Var(\hat{f}(x))$ is high

$$Z_1, Z_2, \dots, Z_m \quad Var(Z_i) = \sigma^2, \quad i = 1, 2, \dots, m$$

$$Var(\bar{Z}) = \frac{\sigma^2}{m} \quad (\text{if } Z_i\text{'s are indep})$$

In order to reduce the variance of the regression tree estimator, we can use the following algorithm.

Data: $\{X_1, X_2, \dots, X_n; y_1, y_2, \dots, y_n\}$

Step 1: Generate a bootstrap sample of size n (A sample of size n is drawn from the data using SRSWR) $\rightarrow$ Simple random sampling with replacement

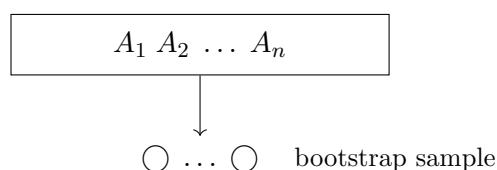


Figure 3: SRSWR - Simple random sampling with replacement

Expected no. of distinct objects in the sample?

Effective sample size:

$$N_0 = \sum_{i=1}^n \mathbf{1}_{\{A_i \text{ is in the sample}\}}$$

$$\begin{array}{l}
 \swarrow \quad \nwarrow \\
 1 \text{ w.p. } 1 - \left(\frac{n-1}{n}\right)^n \\
 0 \text{ w.p. } \left(\frac{n-1}{n}\right)^n
 \end{array}$$

$$E(N_0) = \sum_{i=1}^n E[\mathbf{1}_{\{A_i \text{ is in the sample}\}}] = \sum_{i=1}^n \left\{ 1 - \left(\frac{n-1}{n}\right)^n \right\} = n \left\{ 1 - \left(\frac{n-1}{n}\right)^n \right\}$$

$$E\left(\frac{N_0}{n}\right) = 1 - \left(\frac{n-1}{n}\right)^n$$

$$\begin{aligned}
 \lim_{n \rightarrow \infty} E\left(\frac{N_0}{n}\right) &= \lim_{n \rightarrow \infty} 1 - \left(\frac{n-1}{n}\right)^n = \lim_{n \rightarrow \infty} 1 - \left(1 - \frac{1}{n}\right)^n \\
 &= 1 - \lim_{n \rightarrow \infty} \left(1 - \frac{1}{n}\right)^n = 1 - e^{-1} \approx 0.632 \approx 2/3
 \end{aligned}$$

Let $\{(X_{1i}^{(b)}, Y_{1i}^{(b)}), \dots, (X_{ni}^{(b)}, Y_{ni}^{(b)})\}$ be the sub-sample

Step 2: Based on this sub-sample, grow a large sized tree as before (but do not use pruning).

Let $f^{(b)}$ be the regression tree estimator obtained from this tree.

Repeat Step 1 and Step 2, several times (B times) to get $f^{(b)}$ for $b = 1, 2, \dots, B$ and use $\hat{f}(x) = \sum_{b=1}^B f^{(b)}(x)/B$ as the final estimate for $f(x)$.

This bootstrap aggregation method is known as **Bagging**.

4 Variance of Averaged Estimators

How to choose p' ? How many sub samples? What modifications are needed for classification tree?

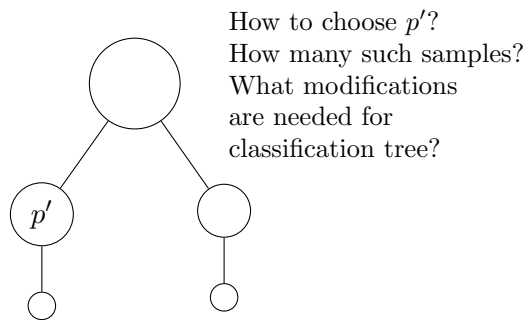
$$Z_1, \dots, Z_m \quad \text{Var}(Z_i) = \sigma^2$$

$$\bar{Z} = \frac{\sum Z_i}{m} \quad \begin{cases} E(Z_1) = \dots = E(Z_m) \\ \downarrow \\ E(\bar{Z}) = E(Z_1) = \dots = E(Z_m) \end{cases}$$

$\text{Var}(\bar{Z}) \approx \frac{\sigma^2}{m}$ if $Z_1, \dots, Z_m$ are independent

$$\begin{aligned}
 \text{Var}(\bar{Z}) &= \frac{1}{m^2} \left[\sum_{i=1}^m \text{Var}(Z_i) + \sum_{i \neq j} \text{Cov}(Z_i, Z_j) \right] \\
 &= \frac{1}{m^2} [m\sigma^2 + m(m-1)\rho\sigma^2]
 \end{aligned}$$

where $\rho = \text{Corr}(Z_i, Z_j)$



5 Aggregation of Trees

Aggregation of trees
 - Bagged regression trees
 - Random forest

5.1 Bagged Regression Tree

1. Generate a sub sample of size n using SRSWR.
2. Construct a regression tree based on that sub sample. (Use the same stopping criterion as before but do not use pruning)

Repeat Steps 1 and 2 several times (B many times)

Let $\hat{f}_1, \hat{f}_2, \dots, \hat{f}_B$ be the B regression tree estimators in these B cases. Then finally we use:

$$\hat{f}_{\text{BAG}}(x) = \frac{1}{B} \left(\hat{f}_1(x) + \dots + \hat{f}_B(x) \right) \text{ as the Bagged regression tree estimator.}$$

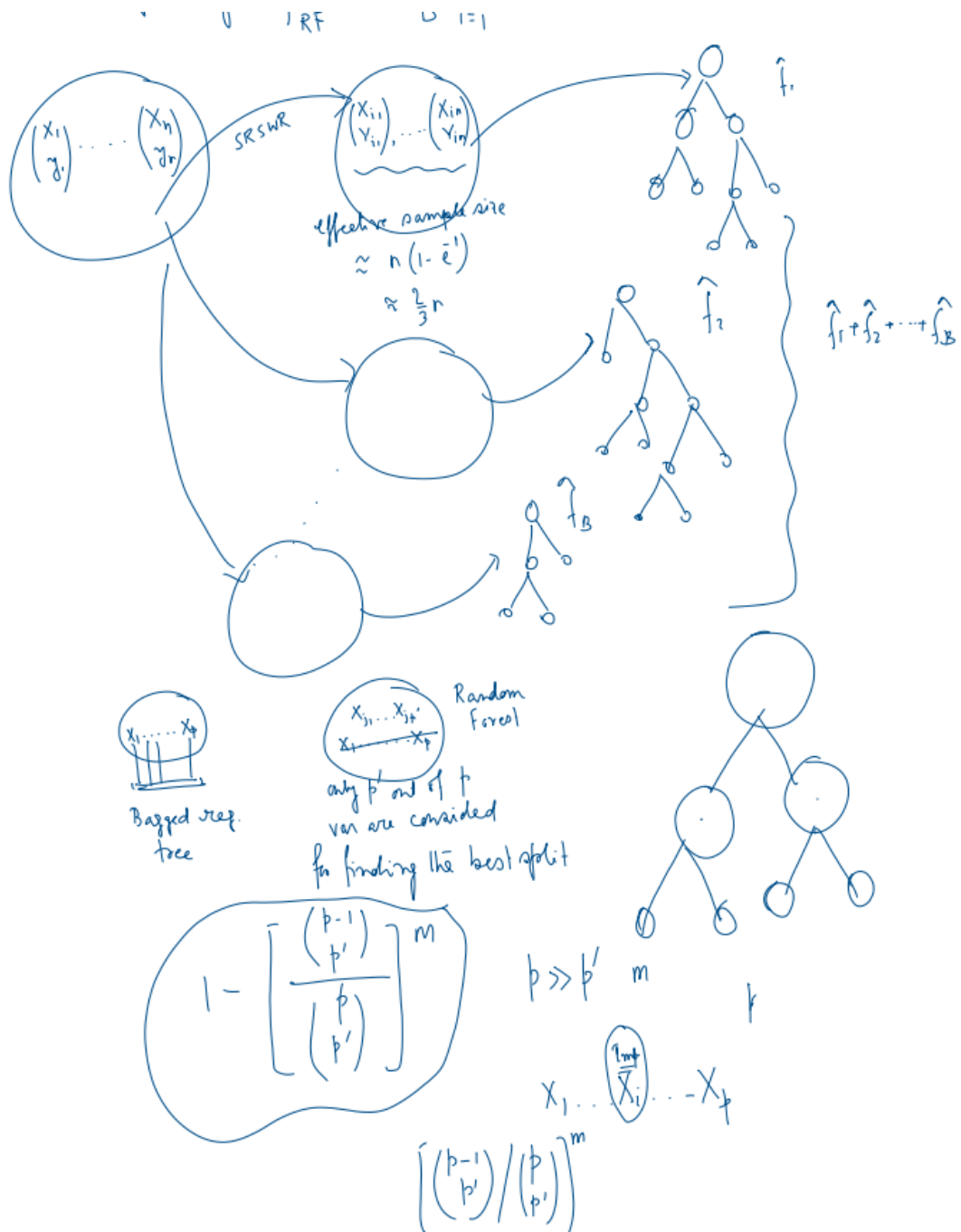
5.2 Random Forest

1. Generate a sub sample of size n using SRSWR.
2. Construct a regression tree based on that sub sample. The same stopping criterion can be used and prior pruning should be avoided. But unlike bagged regression trees, we do not consider all features, we consider only a subset of features at a time to find out the best split (chosen randomly) at a time to find out the best split.

Steps 1 and 2 are repeated several times.

If $\hat{f}_1^*, \hat{f}_2^*, \dots, \hat{f}_B^*$ are the regression tree estimators, then the random forest estimator of the regression function is given by:

$$\hat{f}_{\text{RF}}(x) = \frac{1}{B} \sum_{i=1}^B \hat{f}_i^*(x).$$

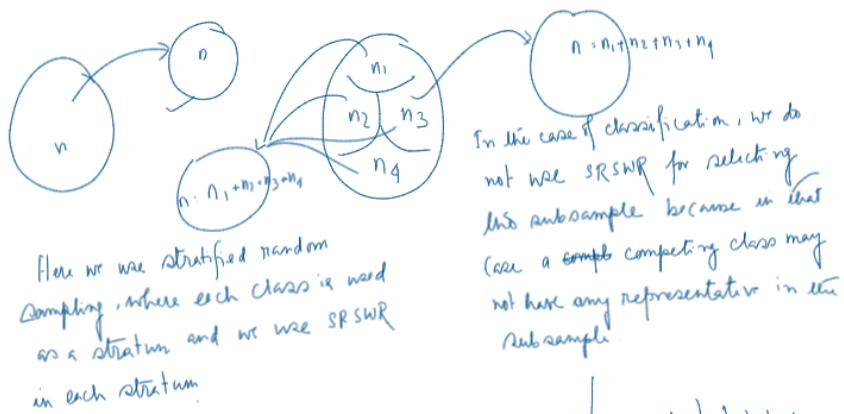
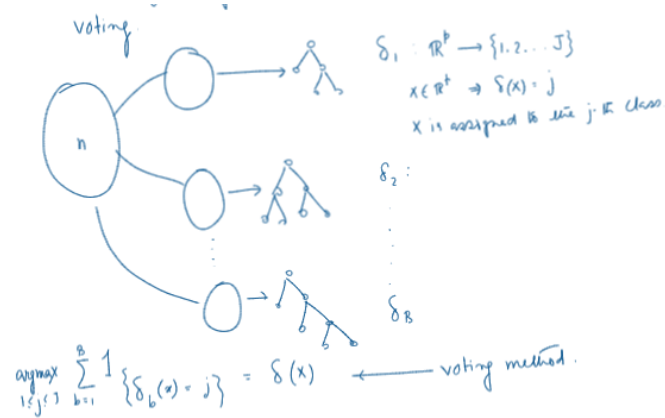


For classification, the procedure remains almost the same. (Sub sample selection & construction of trees are done in the same way) But instead of taking the average at the final step, we use voting.

$$S_i : \mathbb{R}^p \rightarrow \{1, 2, \dots, J\}$$

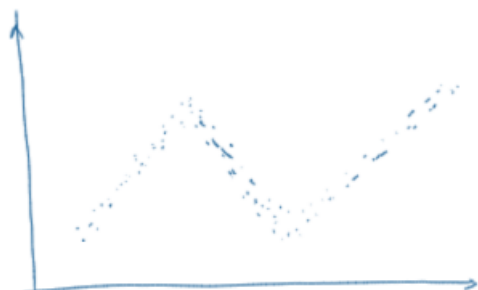
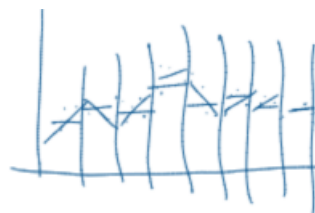
$$x \in \mathbb{R}^p \Rightarrow S(x) = j$$

x is assigned to the j -th class.



in each iteration:

MARS (Multivariable Adaptive Regression Splines)

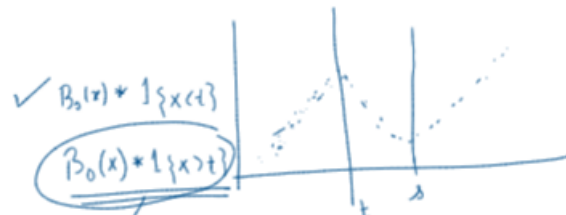
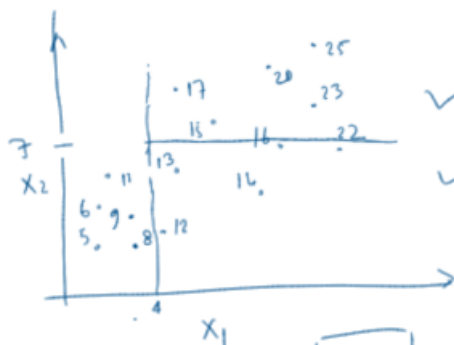


$$B_0(x) = 1$$

$$\sum (y_i - a B_0(x_i))^2$$

$$\sum (y_i - a)^2 \rightarrow a = \bar{y}$$

Regression tree



$$B_0(x) * 1\{x < t\}$$

$$B_0(x) * 1\{x > t\}$$



$$B_0(x) = 1$$

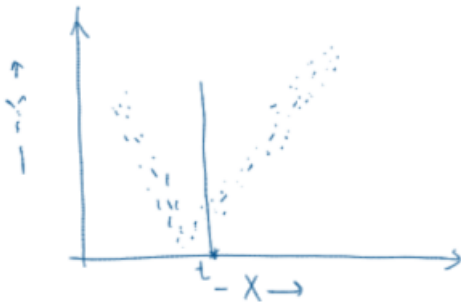
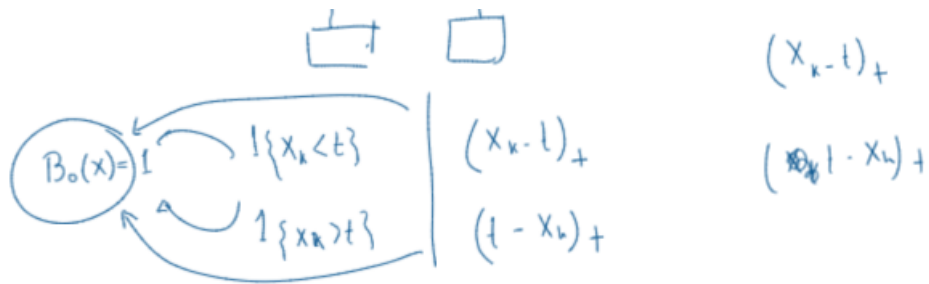
$$B_0(x) * 1\{x_1 > 4\}$$

$$B_0(x) * 1\{x_1 < 4\}$$

$$B_0(x) * 1\{x_1 > 4\} * 1\{x_2 > 7\}$$

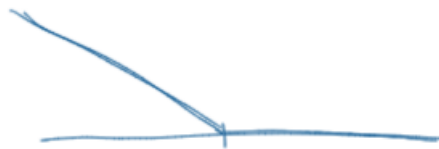
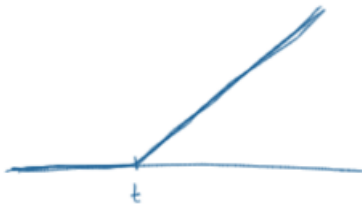
$$B_0(x) * 1\{x_1 > 4\} * 1\{x_2 < 7\}$$

$$(X_k - t)_+$$

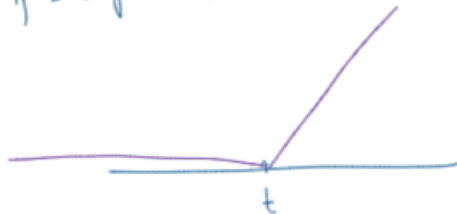


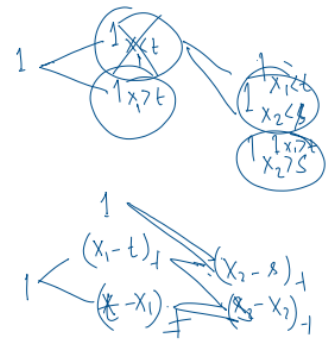
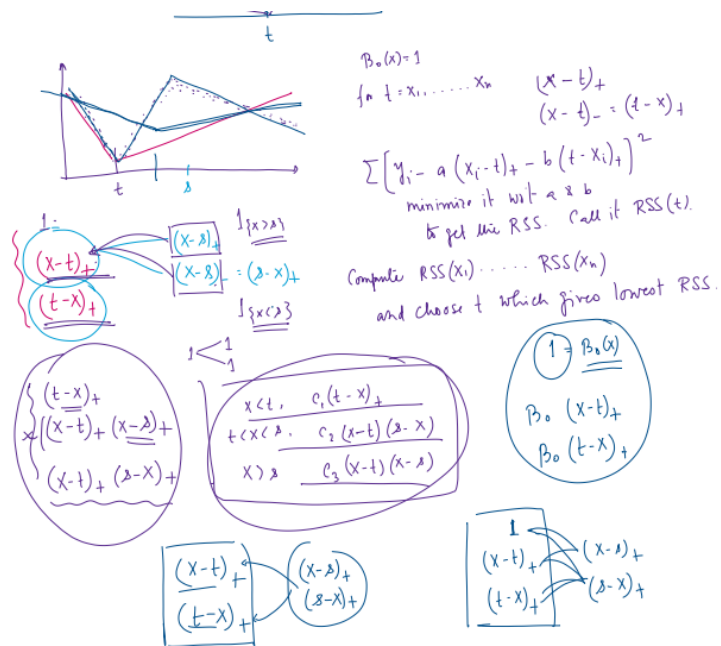
$B_0(x) = 1$
 Reg. based on this gives $\hat{f}(x) = \bar{y}$ for all x .

$$B_0(x) \begin{cases} B_0(x)(x-t)_+ = \begin{cases} x-t & \text{if } x > t \\ 0 & \text{o.w.} \end{cases} \\ B_0(x)(t-x)_+ = \begin{cases} t-x & \text{if } t > x \\ 0 & \text{o.w.} \end{cases} \end{cases}$$



What if I regress Y on these two basis functions?





Start with a basis $B_0 = 1$
 Suppose that after the m th stage, you have basis
 functions $B_0, B_1, B_2, \dots, B_m$.

Choose the var X_i
 Choose a cent t
 $\{x_{i1}, x_{i2}, \dots, x_{in}\}$

$B_0, B_1, \dots, B_m$
 $B_{m+1} = (x_i - t)_+ B_j$
 $B_{m+2} = (t - x_i)_+ B_j$
 $[j = 0, 1, \dots, m]$

$p \times n \times m$
 $(x_{i1} - t)_+, (x_{i2} - t)_+, \dots, (x_{in} - t)_+$
 $(t - x_{i1}), (t - x_{i2}), \dots, (t - x_{in})$